



Association between behavioral factors and recurrence rate in cats with feline "idiopathic" cystitis

Marianne Caudron^a , Pascale Laroche^a, Isabelle Bazin^{a,b}, Marion Desmarchelier^{a,c,*}

^a Centre Hospitalier Universitaire Vétérinaire, Faculté de Médecine Vétérinaire, Université de Montréal, 3200, Sicotte, Saint-Hyacinthe, Québec J2S 2M2, Canada

^b Centre Vétérinaire Daubigny, 3405 Wilfrid-Hamel Blvd, Québec, Québec G1P 2J3, Canada

^c Department of Clinical Sciences, Faculté de Médecine Vétérinaire, Université de Montréal, 3200, Sicotte, Saint-Hyacinthe, Québec J2S 2M2, Canada

ARTICLE INFO

Keywords:

Feline idiopathic cystitis
Internal medicine
Behavior
Anxiety
Recurrences

ABSTRACT

Feline idiopathic cystitis (FIC) is a lower urinary tract disease associated with significant morbidity and mortality in cats. The aim of our study was to detect potential associations between behavioral parameters and recurrence rates in cats diagnosed with FIC. Our main hypothesis was that anxiety-related behaviors could be over-represented in cats that will experience a recurrence, compared to cats with a single episode. For this retrospective study, we recruited cats diagnosed with FIC who presented to the Centre Hospitalier Universitaire Vétérinaire (CHUV) from March 2019 to February 2022. Cats with concurrent systemic diseases or with any other condition that could explain the urinary signs were excluded. Cat owners were sent an online survey to collect data on their cat's behavioral parameters, potential recurrences, and whether they complied or not with provided environmental recommendations. Thirty-three surveys from 23 neutered male and 10 spayed female feline patients were received and analyzed. We found a significant association between fear of unfamiliar people and recurrence of FIC: 94% of cats who had recurrences were reported by their owners to show fear of strangers had recurrences compared to 59% of cats in the single episode group ($P=0.04$). Cats with reported fearful behavior in the presence of familiar people also tended to be overrepresented in the group with recurrences ($P=0.08$). Factors that did not differ between the two groups include displaying aggression towards humans and other animals, being a victim of aggression, displaying urine marking, owner compliance with the prescribed environmental modifications ($P>0.05$). Identifying cats with fear and anxiety-related problems and treating them may help decrease the recurrence rate of FIC and may decrease associated mortality. Further larger scale prospective studies are needed.

Introduction

Feline idiopathic cystitis (FIC) is the most frequent feline lower urinary tract disease (FLUTD) (approximately 55%–65% of total cases of FLUTD (Lekcharoensuk et al., 2001; Sævik et al., 2011)). Although its pathophysiology remains poorly understood (He et al., 2022; Sparkes, 2018), typical clinical signs of FIC include dysuria, hematuria, pollakiuria, stranguria and periuria (Defauw et al., 2011; Westropp et al., 2019). To date, FIC remains a diagnosis of exclusion (Sævik et al., 2011; He et al., 2022), requiring to rule out other differential diagnoses such as urolithiasis, urinary tract infection and neoplasia (Defauw et al., 2011; Westropp et al., 2019). Typical findings on urinalysis of cats with FIC are

sterile urine, absence of urolithiasis and the presence of hematuria and/or proteinuria (Sævik et al., 2011; He et al., 2022). Risk factors that have already been identified include obesity, purebred cats, fearful behavior, and being an indoor cat (Buffington et al., 2006a; Defauw et al., 2011; Lund et al., 2016). Several studies have shown that approximately 40–60% of cats with FIC will experience a recurrence in their life (Markwell et al., 1999; Gunn-Moore and Shenoy, 2004; Defauw et al., 2011; Naarden and Corbee, 2020). Environmental recommendations including dietary changes have been suggested to prevent recurrence of FIC (Buffington et al., 2006b; Westropp et al., 2019; Naarden and Corbee, 2020). Environmental recommendations include an appropriate number of litter boxes, the presence of a private resting area

* Corresponding author at: Centre Hospitalier Universitaire Vétérinaire, Faculté de Médecine Vétérinaire, Université de Montréal, 3200, Sicotte, Saint-Hyacinthe, Québec J2S 2M2, Canada.

E-mail addresses: marianne.caudron@umontreal.ca (M. Caudron), pascale.laroche1@gmail.com (P. Laroche), isabelle.bazin@daubigny.ca (I. Bazin), marion.desmarchelier@umontreal.ca (M. Desmarchelier).

<https://doi.org/10.1016/j.jveb.2025.02.005>

Received 4 September 2024; Received in revised form 11 February 2025; Accepted 17 February 2025

Available online 19 February 2025

1558-7878/© 2025 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

(preferably elevated) and environmental enrichment (e.g., outdoor access, toys in rotation, play). In addition, dietary recommendations include increasing daily water intake (e.g., wet food, water fountain) and feeding specialized urinary diets (Buffington et al., 2006b; Westropp et al., 2019; Naarden and Corbee, 2020). Mortality associated with FIC appear to be high, as approximately 20% of these cats are euthanized 3.5 years after the initial episode according to a recent study (Eggertsdóttir et al., 2021). Feline idiopathic cystitis affects the quality of life of many cats, being associated with substantial pain and discomfort (Buffington, 2004). To this date, risks factors associated with the high recurrence rate of FIC have not been well studied.

Though the pathophysiology of FIC remains largely unknown, several recent studies have shown that FIC is likely a psychoneuroendocrine pathology (Buffington et al., 1996b, 2002; Reche Jr and Buffington, 1998; Westropp et al., 2003; He et al., 2022;). Studies showed that cats with FIC may have an exaggerated adrenergic stimulation combined with an inappropriate adrenal response (Buffington et al., 2002; Reche Jr and Buffington, 1998). First, the exaggerated adrenergic stimulation may cause damage to tight junctions of the bladder cells, exposing sensory neuron called C-fibers (pain sensors located in the bladder submucosa) (Buffington et al., 1996b; Lavelle et al., 2000). As a result, the production of substance P may contribute to pain, vasodilation, and increased bladder wall permeability and increased mast cell degranulation (Pang et al., 1995; Buffington and Wolfe Jr., 1998; He et al., 2022). In addition, cats with FIC may also have an inappropriate adrenal response. In cats with normal adrenal function, cortisol creates negative feedback on the sympathetic adrenal medullary system, decreasing adrenergic activity (He et al., 2022). However, Westropp et al. (2003) found that adrenal glands of cats with FIC were smaller, and did not secrete enough cortisol when stimulated by ACTH, allowing for continued adrenergic stimulation. Furthermore, cats with FIC also appear to have an abnormal bladder barrier composition (Buffington et al., 1996a) and decreased urinary excretion of glycosaminoglycans (GAGs), which are polysaccharides present in the bladder wall providing an important defense mechanism for the urothelium. These GAGs interact with urine and protect the urothelium from urinary irritants (Buffington et al., 1996a; He et al., 2022). If cats with FIC have fewer GAGs on their bladder wall, the urothelium is less protected from urinary irritants which can result in urothelium ulceration (Lavelle et al., 2000). Ultimately, damage to the urothelium stimulates the release of substance P. As a result, urinary irritants coupled with an exaggerated adrenergic stimulation that leads to the production of substance P may induce neurogenic inflammation in the bladder of cats with FIC (Buffington et al., 1996b; Buffington and Wolfe Jr., 1998; He et al., 2022).

As a psychogenic component could be involved in this complex process, the aim of our study was to detect a potential association between measured behavioral parameters (presence of fear, aggression and urine marking, as reported by the owners) and the recurrence rates in cats diagnosed with FIC. Additional objectives were to evaluate the potential impact of environmental recommendations and feeding patterns (*ad libitum* or meals) on the recurrence rates of FIC.

Our main hypothesis was that anxiety-related behaviors would be overrepresented in cats that experienced a recurrence, compared to cats with a single episode. We also hypothesized that reported compliance to environmental recommendations would be associated with a lower recurrence rate. Finally, we hypothesized that a portion-controlled feeding pattern would be associated with a higher recurrence rate than *ad libitum* feeding.

Material and methods

Inclusion criteria

For this retrospective study, we recruited cats that were presented to the veterinary teaching hospital, Centre Hospitalier Universitaire

Vétérinaire (CHUV), between March 2019 and February 2022 and had been diagnosed with FIC based on clinical signs (hematuria, dysuria, stranguria and periuria), urology, and imaging (radiographs and/or ultrasound). Feline idiopathic cystitis diagnosis for inclusion in our study was made based on typical urinary clinical signs in the absence of bacteriuria and pyuria, as well the absence of urolithiasis or vesical mass on diagnostic imaging. Cats with a concurrent systemic disease or with any other condition that could explain the observed urinary signs (including urethral obstruction and urolithiasis) were excluded from this study. Cats that were euthanized after their diagnosis were also excluded. Cats with inactive chronic conditions unrelated to the urinary tract system (i. e. cardiomyopathy) were included in the study.

Case management and standardized discharge recommendations

All cats were admitted by a veterinarian at the teaching hospital (CHUV) and received the highest standards of care based on each individual clinical assessment. Owners all received specific environmental and dietary recommendations upon discharge from the hospital (see Table 1). Standard hospital treatment protocol for all cats with FIC included prescriptions of gabapentin (6–12 mg/kg PO q. 8–12h for 5–14 days), buprenorphine (0.009–0.019 mg/kg PO q. 8–12h for 3–10 days), and prazosin (0.06–0.40 mg/kg PO q. 12h for 5–10 days). One cat included in the study received only a prescription of a non-steroidal anti-inflammatory drug, meloxicam (0.05mg/kg PO q. 24h for 3 days). Adverse effects of each medication were explained to owners. Owners were also informed to watch for specific clinical signs following the veterinary visit: presence of blood in the urine, signs of pain during urination (vocalization during urination, urinating outside the litter box), abdominal efforts during urination and drip-like urination.

Table 1
Specific recommendations provided to the owners during the veterinary visit (based on Buffington et al., 2006b; Westropp et al., 2019; Naarden and Corbee, 2020).

Categories	Recommendations
Water intake	• Wet food or adding water to dry food. • Several bowls of water distributed throughout the house. • Fresh water always available. • Wide bowls for cats who do not like their whiskers touching the bowl. • Food and water bowls kept in a quiet room. • Ice cubes made of fish or meat stock in the cat's water. • Water fountain.
Diet change	• Specialized commercially available urinary diet
Litter boxes	• Strict litter box hygiene: owners should empty the excrements every day and clean the box once a week with a mild soap. • Provide at least one more litter box than the number of cats in the house. • Provide one litter box on each floor. • Cats should choose the substrate, the box size and the type of litter (covered or not). • Non-perfumed clumping litter seems to be preferred by cats.
Stress reduction/ environmental enrichment	• Bowls and litter boxes kept in a quiet room away from disturbance (e.g., children, other animals, noisy appliances, etc.). • Cats should have a private resting area where they feel safe and where they can observe outside and their environment. This resting area should preferably be elevated (e.g., hammock, windowsill). • Owners should provide a claw station and offer various toys in rotation, and imaginary predation activity (e.g., laser light, hiding food in the house, play time, etc.). • Consider the possibility of giving the cat safe outdoor access (e.g., with a harness, a catio, etc.).

Owners were also notified to visit a veterinarian as soon as possible if their cat did not urinate for a period of 12–24 h.

Survey

Cat owners were emailed an online survey to collect data on their cat's behavior (presence of fear, aggression, victim of aggression, urine marking), potential recurrences, whether they complied or not with environmental recommendations provided after the first episode of FIC and feeding patterns (see Table 2). Questions and behavioral descriptions were kept short to increase the response rate and allow for future use of these questions in an emergency setting.

Statistical analyses

Descriptive data are presented as counts and percentages. Data regarding the effect of different behavioral parameters on the recurrence or not of FIC was analyzed with generalized linear models. The dependent variable was binary (recurrence of cystitis: 1; no recurrence: 0). To increase the statistical power, for all behavior-related questions, we combined the answers “Sometimes” and “Often” as opposed to the answers “Never” to obtain only two options: the behavior was either reported or not. Data distribution was binomial. A Chi-square test and a Fisher's exact test, when the conditions for using a Chi-square test were not met, were used to assess the association between each parameter and the recurrence or not of FIC. Student *t*-test was also used to establish the effect of the weight and age on the recurrence or not of FIC. A P-value < 0.05 was considered statistically significant. Statistical analyses were conducted using R (version 4.3.1).

Results

Sixty cats were included in the study and 50 surveys were sent to owners who gave their consent to participate in the study. We received 35 surveys (response rate of 70%). Two surveys were excluded, one due to missing information on recurrences and the other due to a reported history of pyelonephritis after the initial consultation for FIC. We therefore analyzed 33 surveys from 23 neutered males and 10 spayed females. Patients were then separated in two groups: cats with a single episode ($n = 17$) and cats with two or more episodes ($n = 16$). Average time between the first FIC episode and time of survey was 21.44 +/- 12.35 months (min: 8.74 months; max: 60.30 months; median: 18.41 months; $n = 33$ cats). 16 cats had already had a recurrence within this period. The remaining 17 cats who had not recurred at the time of the first survey were contacted again 1 year and 2 years after the initial survey to document potential recurrences over a longer period of time. During the one-year follow-up, only 12 out of the 17 owners answered the survey (loss of follow-up = 5 cats). During this period, two cats from the single episode group were deceased of a cause unrelated to FIC. Therefore, there were 15 cats in the single episode group and 16 cats in the recurrence group (2 or more episodes). During the two-year follow-up, 8 out of 15 owners answered the survey (loss of follow-up = 7 cats, the same 5 cats as at the one-year follow-up plus 2 new cats), resulting in a follow-up time of an average 28.63 +/- 13.40 months (min: 8.74 months; max: 48.56 months; median: 29.42 months; $n = 17$ cats). After the one and two-year follow-up, no additional cat had recurred in the single episode group. The recurrence rate was 48% at the time of survey (21.44 +/- 12.35 months after the first FIC episode).

Two cats received a fluoxetine treatment before they were presented at the CHUV for their FIC episode. One cat from the recurrence group had been on fluoxetine for over 3 years for spraying behavior. The other cat was in the single episode group and had been on fluoxetine for a behavioral problem for an unknown period of time. The nature of the behavioral disease was not reported in the available medical records. None of the cats had consulted a veterinary behaviorist before the survey was sent.

Table 2

Online survey sent to the owners of all selected cats.

Do you consent to the use of the information you will provide in this survey for research purposes? All data will remain anonymous and confidential.	A) Yes B) No
During your visit to the CHUV, when your cat was diagnosed with feline idiopathic cystitis, to your knowledge, did you obtain recommendations to prevent recurrences?	A) Yes B) No
Have you followed these recommendations?	A) Yes B) No C) Partially
Indicate the changes that you have made.	A) Diet change. B) Increasing in the number of litter boxes. C) Change of litter boxes for larger or uncovered boxes. D) Change in the litter substrate. E) Adding play time with your cat. F) Increasing water intake (e.g., water fountain, many water bowls in different places in the house).
Has your cat had any other episodes of feline idiopathic cystitis after the first episode? List those episodes with approximate dates (months and years).	A) Yes B) No
Environment	
What type of housing do you have?	A) Single family home B) Apartment
Does your cat have access to outside?	A) Yes B) No
Do you live with children?	A) Yes B) No
Has the pandemic caused a significant change in routine (e.g., telework, homeschooling for children)?	A) Yes B) No
Did you move in last two years?	A) Yes B) No
Cat	
Do you have other cats?	A) Yes, one more. B) Yes, more than one. C) No
Do you have other animals?	A) Yes, one/many dogs. B) Yes, another species. C) No
What type of litter do you currently use?	A) Clumping B) Non-clumping
How many litter boxes do you have?	
How is your cat fed?	A) <i>Ad libitum</i> (food distributed as often as desired) B) Per meal with periods of time when food is not accessible.
Does your cat have a place where he can rest without being disturbed?	A) Yes B) No
Behavior	
Does your cat exhibit fearful behavior in the presence of familiar people (e.g., hiding, jumping, etc.)?	A) Often B) Sometimes C) Never
Does your cat exhibit fearful behavior in presence of unfamiliar people (e.g., hiding, jumping, etc.)?	A) Often B) Sometimes C) Never
Does your cat show aggression towards familiar people?	A) Often B) Sometimes C) Never
Does your cat show aggression towards unfamiliar people?	A) Often B) Sometimes C) Never
Does your cat show aggression towards other cats (e.g., active attack, stares at another cat for a long time)?	A) Often B) Sometimes C) Never D) N/A
Does your cat show aggression towards other animals?	A) Often B) Sometimes C) Never

(continued on next page)

Table 2 (continued)

	D) N/A
Is your cat being attacked by another animal?	A) No B) Yes, by another cat. C) Yes, by another animal.
Does your cat ever urinate out of the litter box? (e.g., urinary marking, spray on vertical surfaces)?	A) Often B) Sometimes C) Never
Comment (e.g., issue, euthanasia, other health problem).	Open-ended question

Breeds

Twenty-five cats out of 33 were domestic cats, 3 were Bengals, 1 was a Cornish Rex, 1 was a Highland Lynx, 1 was a Persian, 1 was an Abyssinian and 1 was a Siberian.

Weight and age

No statistical differences were observed between the weight and age and the recurrence or not of FIC (P= 0.17 and P=0.96) (Table 3).

Behavioral parameters

From the survey sent to the owners, we were able to analyze the following parameters (as reported by the owners): fearful behavior in the presence of familiar people, fearful behavior in the presence of unfamiliar people, aggression towards humans, aggression towards other animals, victim of aggression, urine marking, owner compliance with the prescribed environmental modifications and method of food delivery (see Tables 4 and 5). The other parameters were not observed frequently enough or had a distribution that did not allow for statistical analysis with the low number of patients in this study.

Cats reported with fear of unfamiliar people were significantly more at risk to experience a recurrence after their first episode of FIC both with the generalized linear models (when data were separated between when the behaviors were reported to occur “often”, “sometimes”, or “never” (the last one being used as the intercept; see Table 4) and with the Fischer’s exact test (with data combined in 2 categories whether the behavior either occurred or not (grouping data from “often” and “sometimes” categories versus “never”)) (P=0.04). More specifically, in the single episode group, 10 cats out of 17 were reported to be fearful in presence of unfamiliar people (59%) while in the recurrence group, 15 cats out of 16 were fearful of strangers (94%). Furthermore, cats reported with fearful behavior in the presence of familiar people tended to be overrepresented in the group with recurrence (13 cats out of 16; 81%) compared to the single episode group (9 cats out of 17; 53%) ($X^2 = 2.97$; P = 0.08). House soiling, aggression towards humans and other animals, being a victim of aggression, as well as the method of food delivery and owner compliance to environmental modifications, did not differ between the 2 groups (Tables 4 and 5).

Table 3

Breeds, mean and median of weight (kg) and age at the time of survey (years) of cats in each group (single episode and recurrence). A Student t-test was used for statistical analysis.

		Single episode		Recurrence (2 or more episodes)		T-value	df	SE	P
Breeds		15 domestic cats, 1 Bengal and 1 Cornish Rex		10 domestic cats, 2 Bengals, 1 Highland lynx, 1 Persian, 1 Abyssinian and 1 Siberian					
Weight (kg)	Mean	5.8 ± 1.5		5.1 ± 1.2		1.41	31	0.48	0.17
	(± SD)								
Age (years)	Median	5.4		5.5		0.04	31	1.14	0.96
	Mean	4.3 ± 3.4		4.3 ± 3.1					
	(± SD)								
	Median	3.2		3.9					

Table 4

Results of generalized linear models (GLMs) on the effect of different behavioral and environmental parameters on cystitis recurrence (yes or no). Significant results are in bold.

Parameters	Categories	Estimate (± SE)	Z- score	P- value
Fearful behavior in presence of familiar people	Intercept	−0.98 (0.68)	−1.45	0.147
	Sometimes	1.39 (0.86)	1.62	0.106
	Often	1.27 (1.02)	1.24	0.214
Fearful behavior in presence of unfamiliar people	Intercept	−1.95 (1.07)	−1.82	0.069
	Sometimes	2.64 (1.28)	2.06	0.040
	Often	2.20 (1.18)	1.86	0.063
Aggression towards humans	Intercept	0.40 (0.10)	4.07	< 0.001
	Sometimes	0.35 (0.20)	1.75	0.090
Aggression towards other animals	Intercept	−0.81 (0.60)	−1.35	0.177
	Sometimes	1.22 (0.88)	1.38	0.168
	Often	0.81 (1.17)	0.70	0.487
Victim of aggression	Intercept	0.00 (0.00)	0.00	1.000
	Yes	0.00 (0.00)	0.00	1.000
House soiling (e.g., urine marking)	Intercept	0.00 (0.35)	0.00	1.000
	Yes	−16.57 (2400.00)	−0.01	0.994
Compliance with the prescribed environmental modifications	Intercept	−17.57 (2797.44)	−0.01	0.995
	Yes	18.01 (2797.44)	0.01	0.995
	Partially	16.47 (2797.44)	0.01	0.995
Type of feeding	Intercept	0.29 (0.44)	0.65	0.514
	Ad libitum	−0.98 (0.76)	−1.30	0.194

Discussion

The recurrence rate of FIC in the current study was 48% at the time we received the survey (within an average of 21.44 +/- 12.35 months) and did not appear to increase over the following 2 years in the surviving cats. These results are in agreement with two other studies where the recurrence rate was 39% (in the United States) (Buffington et al., 2006b) and 51.6% within a year (in the Netherlands) (Naarden and Corbee, 2020). In another study from Scotland, FIC recurrence rate was reported to be as high as 65% within six months (Gunn-Moore and Shenoy, 2004). This high recurrence rate demonstrates that we are currently unable to prevent recurrences in half of our FIC patients, despite provision and reported good compliance of current literature recommendations.

Though not a goal of our study, we did not observe that being a pure bred cat was a risk factor for recurrence of FIC. However, the small number of cats in our study limits this interpretation. Previous findings showed that Persian cats and domestic cats were more at risk of presenting FIC (Cameron et al., 2004, Lund et al., 2016), but risks of recurrences were not studied.

According to the literature, cats with FIC are typically between 2 and 7 years old (Cameron et al., 2004; Gunn-Moore, 2003; Lekcharoensuk et al., 2001). Similarly to our results, where the mean age of single episode group and recurrence group were 4.3 +/- 3.3 and 4.3 +/- 3.1 respectively, the mean age of cats with FIC in a study was 3.8 +/- 2.6 years (Markwell et al., 1999). In a recent study, cats diagnosed with FIC

Table 5

Association between the occurrence of FIC recurrence and behavioral and environmental parameters after grouping results by whether the behavior was reported (“yes” = categories “Often” and “Sometimes” or “Yes” and “Partially” seen in Table 4) or not (no = category “Never” in Table 4). *Chi-square test or Fisher’s exact test were used to obtain the different p-value.*

Parameters	Group	No (n of cats)	Yes (n of cats)	P
Fearful behavior in presence of familiar people	Single episode	8	9	0.08
	Recurrence	3	13	
Fearful behavior in presence of unfamiliar people	Single episode	7	10	0.04
	Recurrence	1	15	
Aggression towards humans	Single episode	15	2	>0.10
	Recurrence	10	6	
Aggression towards other animals	Single episode	9	6	>0.10
	Recurrence	4	8	
Victim of aggression	Single episode	13	3	>0.10
	Recurrence	13	3	
House soiling (e.g., urine marking)	Single episode	16	1	>0.10
	Recurrence	16	0	
Compliance with the prescribed environmental modifications	Single episode	2	15	>0.10
	Recurrence	0	16	
Type of feeding		Ad libitum	Meals	>0.10
	Single episode	8	9	
	Recurrence	4	12	

were mostly seen in younger cats, while older cats (> 10 years) were mostly diagnosed with neoplasia and urinary tract infections (Lew-Kojrys et al., 2017).

Several studies have reported that fearful behavior was a risk factor for FIC (Buffington et al., 2006a; Defauw et al., 2011; Lund et al., 2016). Our results showed that cats reported with fearful behavior in the presence of unfamiliar people had a higher risk of recurrence after their initial episode of FIC. Fear of familiar people was also reported more often in the FIC recurrence group, but this difference was not statistically significant. These behavioral parameters should be investigated in future prospective studies as our small sample size may have led to a type II error. However, it is important to mention that cats presenting with fear of familiar people likely experience chronic stress and would benefit from a behavioral consultation to improve their health and welfare.

A previous study have shown that aggressive behavior was not associated with a higher prevalence of FIC (Defauw et al., 2011; Lund et al., 2016). However, two other studies suggested the contrary. The first one showed that cats with aggressive behavior were over-represented in cats with clinical signs of LUTD (Buffington et al., 2006a), and the second study showed that inter-cat conflict in the same household was associated with FIC (Cameron et al., 2004). In our study, no statistical difference were observed between cats who displayed aggression (towards humans and animals) and the recurrence rate of FIC. In the literature, there is a statistical tendency for cats who are victims of aggression (described as “threatened”) to develop FIC in situations of silent inter-cat conflicts (Defauw et al., 2011; Westropp et al., 2019). A silent inter-cat conflict is often subtle with no vocalizations and can easily be missed by owners. It may include piloerection, staring at the other cat, avoiding contacts, spraying behavior and other subtle signs (Westropp et al., 2019). However, in our study, cats who were reported as victims of aggression were not overrepresented, as for cats who displayed aggression, in the group with recurrence, though our small sample size and the fact that we based our questionnaire on owners’ observations could again limit this interpretation. The “victim of aggression” and “aggression” parameters should be investigated in future prospective studies on FIC recurrences. Furthermore, it is important to note that aggression and fear are not mutually exclusive categories. Cats often display fear-based aggression.

Urine mar Pryor et al., 2001 king, especially in spayed or neutered animals, is a well-recognized sign of feline anxiety (Pryor et al., 2001; Hart et al., 2005; Levine, 2008). Westropp et al. in 2019 showed that both “assertive” and “threatened” cats (as defined in their publication) can exhibit conflict-related urine marking, but that FIC occurs mostly in

“threatened” cats (Defauw et al., 2011; Westropp et al., 2019). In our study, only one cat was reported to display spraying behavior preventing further analysis of urine marking as a risk factor for FIC recurrence. Cameron et al. (2004) did not find any statistical difference for the frequency of urine marking between the FIC group and the control group. Nevertheless, this parameter should still be considered in future surveys assessing links between FIC and feline anxiety.

In the present study, implementation of environmental modifications by the owners were reportedly similar in both groups. This outcome differs from another study that suggests that increasing water intake reduces the risk of FIC development (Defauw et al., 2011). On the other hand, our results agree with the same study where authors did not find statistical difference between the prevalence of FIC and the number of litter boxes, the location of the litter boxes in the house (quiet room or not) and the number of cats present in the house (Defauw et al., 2011). In our study, almost all owners (31 out of 33) reportedly followed the recommendations. However, we were unable to confirm which recommendations had been followed and for how long. We hypothesize that the recurrence rate may have been higher if the recommendations had not been implemented. Our findings should not be interpreted as undermining the importance of environmental recommendations but show that they may not be sufficient in all cases. Individually-designed environmental recommendations and behavior modification protocols by veterinary behaviorists would deserve further investigation as they may be more efficient than general recommendations to prevent recurrences, but may not be practical in all cases.

A previous study showed that cats with FIC were significantly more frequently fed with portioned meals (36%) compared to cats in the control group who were fed mostly *ad libitum* (81%) ($P=0.016$) (Lund et al., 2016). However, our study did not find any statistical association between method of food delivery and FIC recurrence. Food access limitation can be stressful for cats and can induce inter-cat conflict as it is not a natural feeding pattern (Lund et al., 2016; Westropp et al., 2019). It would be interesting to see the association between food delivery and recurrence rate in prospective studies with a larger sample.

Our study is limited by a small sample size and by its retrospective nature. The diagnostic criteria could be another source of limitation as point-of-care ultrasound was considered a sufficient imaging examination, and no urine culture was requested during the veterinary visit when no bacteriuria and pyuria were seen on cytology. Furthermore, we established associations which should not be confused with causal links between the behavioral parameters studied and the recurrence rate. A recruitment bias was present as our sample came from a single teaching hospital, and we used owner recollection and interpretation for our data

collection. Using a non-validated survey of cat owners is an additional limitation of our study. We intentionally reduced the number and length of the questions as the long-term goal would be to use these findings to detect patients that would be more at risk of recurrences during their first emergency visit. This led to a lack of detail in what behaviors could be included by the owners under the fear or aggression categories. Our categories were also not mutually exclusive as fear aggression is common in cats.

Conclusion

In summary, our results support the hypothesis that cats with untreated fear of unfamiliar people may be at a higher risk of recurrence after their first episode of FIC. Identifying and treating cats with anxiety-related signs may help decrease the recurrence rate of FIC and may decrease associated mortality. This study may help veterinarians decide which patients diagnosed with FIC would most likely benefit from a behavioral medicine consultation by asking simple questions to owners. Larger scale prospective studies are needed, including the use of specific semi-objective questions with a validated questionnaire and the follow-up of FIC patients after a behavioral medicine consultation. As we advance our understanding of psychoneuroendocrine diseases such as FIC, the need to combine mental and physical health assessments to improve the global health of our feline patients becomes more and more obvious and should ultimately improve feline welfare.

Ethical statement

After evaluation of the project prior to the study, no ethical certificate was deemed necessary by the Ethical Comity of Animal Use (Faculté de médecine vétérinaire, Université de Montréal) nor by the Ethical Comity of Health and Science Research (Université de Montréal) due to its retrospective nature.

CRediT authorship contribution statement

Caudron Marianne: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation. **Desmarchelier Marion:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Laroche Pascale:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Data curation. **Bazin Isabelle:** Writing – review & editing, Methodology, Conceptualization.

Conflict of Interest

No conflict of interest to disclose.

Acknowledgments

Tristan Juette, PhD, for the statistical analyses and Yasmeen Prud'homme for her thorough revision of the manuscript. Marianne Caudron received a Veterinary Student Scholarship from the Morris Animal Foundation to support this research project. Dr. Desmarchelier's research program is funded by the National Sciences and Engineering Research Council of Canada (Grant # RGPIN-2023-03848).

Authorship statement

The original idea was conceived by Marion Desmarchelier. The experiment was designed by Pascale Laroche, Isabelle Bazin and Marion Desmarchelier. The data was collected by Pascale Laroche and Marianne Caudron, and analyzed by Pascale Laroche, Marianne Caudron and Marion Desmarchelier. The paper was written by Marianne Caudron and Marion Desmarchelier, and revised by Pascale Laroche and Isabelle

Bazin.

Ethical considerations

After evaluation of the project prior to the study, no ethical certificate was deemed necessary by the Ethical Comity of Animal Use (Faculté de médecine vétérinaire, Université de Montréal) nor by the Ethical Comity of Health and Science Research (Université de Montréal) due to its retrospective nature.

References

- Buffington, T.C., 2004. Comorbidity of interstitial cystitis with other unexplained clinical conditions. *J. Urol.* 172, 1242–1248. <https://doi.org/10.1097/01.ju.0000137953.49304.6c>.
- Buffington, T.C., Blaisdell, L., Binns, P., Woodworth, B.E., 1996a. Decreased urine glycosaminoglycan excretion in cats with interstitial cystitis. *Am. Urol. Assoc.* 155, 1801–1804.
- Buffington, T.C., Dennis, J.C., DiBartola, S.P., 1996b. Interstitial cystitis in cats. *Vet. Clin. North Am. Small Anim. Pract.* 26, 317–326. [https://doi.org/10.1016/S0195-5616\(96\)50212-3](https://doi.org/10.1016/S0195-5616(96)50212-3).
- Buffington, T.C., Teng, B., Somogyi, G.T., 2002. Norepinephrine content and adrenoceptor function in the bladder of cats with feline interstitial cystitis. *Am. Urol. Assoc.* 167, 1876–1880.
- Buffington, T.C., Westropp, J.L., Chew, D.J., Bolus, R.R., 2006a. Risk factors associated with clinical signs of lower urinary tract disease in indoor-housed cats. *J. Am. Vet. Med. Assoc.* 228, 722–725. <https://doi.org/10.2460/javma.228.5.722>.
- Buffington, T.C., Westropp, J.L., Chew, D.J., Bolus, R.R., 2006b. Clinical evaluation of multimodal environmental modification (MEMO) in the management of cats with idiopathic cystitis. *J. Feline Med. Surg.* 8, 261–268. <https://doi.org/10.1016/j.jfms.2006.02.002>.
- Buffington, T.C., Wolfe Jr, S.A., 1998. High affinity binding sites for [(3) H]substance P in urinary bladders of cats with interstitial cystitis. *Am. Urol. Assoc.* 160, 605–611.
- Cameron, M.E., Casey, R.A., Bradshaw, J.W.S., Waran, N.K., Gunn-Moore, D.A., 2004. A study of environmental and behavioural factors that may be associated with feline idiopathic cystitis. *J. Small Anim. Pract.* 45, 144–147. <https://doi.org/10.1111/j.1748-5827.2004.tb00216.x>.
- Defauw, P.A., Van De Maele, I., Duchateau, L., Polis, I.E., Saunders, J.H., Daminet, S., 2011. Risk factors and clinical presentation of cats with feline idiopathic cystitis. *J. Feline Med. Surg.* 13, 967–975. <https://doi.org/10.1016/j.jfms.2011.08.001>.
- Eggertsdóttir, A.V., Blankvandsbråten, S., Gretarsson, P., Olofsson, A.E., Lund, H.S., 2021. Retrospective interview-based long-term follow-up study of cats diagnosed with idiopathic cystitis in 2003–2009. *J. Feline Med. Surg.* 23, 945–951. <https://doi.org/10.1177/1098612X21990302>.
- Gunn-Moore, D.A., 2003. Feline lower urinary tract disease. *J. Feline Med. Surg.* 5, 133–138. [https://doi.org/10.1016/S1098-612X\(02\)00129-8](https://doi.org/10.1016/S1098-612X(02)00129-8).
- Gunn-Moore, D.A., Shenoy, C.M., 2004. Oral glucosamine and the management of feline idiopathic cystitis. *J. Feline Med. Surg.* 6, 219–225. <https://doi.org/10.1016/j.jfms.2003.09.007>.
- Hart, B.L., Cliff, K.D., Tynes, V.V., Bergman, L., 2005. Control of urine marking by use of long-term treatment with fluoxetine or clomipramine in cats. *J. Am. Vet. Med. Assoc.* 226, 378–382. <https://doi.org/10.2460/javma.2005.226.378>.
- He, C., Fan, K., Hao, Z., Tang, N., Li, G., Wang, S., 2022. Prevalence, risk factors, pathophysiology, potential biomarkers and management of feline idiopathic cystitis: an update review. *Front. Vet. Sci.* 9, 900847. <https://doi.org/10.3389/fvets.2022.900847>.
- Lavelle, J.P., Meyers, S.A., Ruiz, W.G., Buffington, T.C., Zeidel, M.L., Apodaca, G., 2000. Urothelial pathophysiological changes in feline interstitial cystitis: a human model. *Am. J. Physiol. Ren. Physiol.* 278, F540–F553. <https://doi.org/10.1152/ajprenal.2000.278.4.F540>.
- Lekcharoensuk, C., Osborne, C.A., Lulich, J.P., 2001. Epidemiologic study of risk factors for lower urinary tract diseases in cats. *J. Am. Vet. Med. Assoc.* 218, 1429–1435. <https://doi.org/10.2460/javma.2001.218.1429>.
- Levine, E.D., 2008. Feline fear and anxiety. *Vet. Clin. North Am. Small Anim. Pract.* 38, 1065–1079. <https://doi.org/10.1016/j.cvsm.2008.04.010>.
- Lew-Kojrys, S., Mikulska-Skupien, E., Snarska, A., Krystkiewicz, W., Pomianowski, A., 2017. Evaluation of clinical signs and causes of lower urinary tract disease in Polish cats. *Veter. Med.* 62, 386–393. <https://doi.org/10.17221/170/2016-VETMED>.
- Lund, H.S., Sævik, B.K., Finstad, Ø.W., Grøntvedt, E.T., Vatne, T., Eggertsdóttir, A.V., 2016. Risk factors for idiopathic cystitis in Norwegian cats: a matched case-control study. *J. Feline Med. Surg.* 18, 483–491. <https://doi.org/10.1177/1098612X15587955>.
- Markwell, P., Buffington, T.C., Chew, D.J., Kendall, M., Harte, J., DiBartola, S., 1999. Clinical evaluation of commercially available urinary acidification diets in the management of idiopathic cystitis in cats. *J. Am. Vet. Med. Assoc.* 214, 361–365.
- Naarden, B., Corbee, R.J., 2020. The effect of a therapeutic urinary stress diet on the short-term recurrence of feline idiopathic cystitis. *Vet. Med. Sci.* 6, 32–38. <https://doi.org/10.1002/vms3.197>.
- Pang, X., Marchand, J., Sant, G.R., Kream, R.M., Theoharides, T.C., J., 1995. Increased number of substance P positive nerve fibres in interstitial cystitis. *Br. J. Urol.* 75, 744–750. <https://doi.org/10.1111/j.1464-410X.1995.tb07384.x>.

- Pryor, P.A., Hart, B.L., Bain, M.J., Cliff, K.D., 2001. Causes of urine marking in cats and effects of environmental management on frequency of marking. *J. Am. Vet. Med. Assoc.* 219, 1709–1713. <https://doi.org/10.2460/javma.2001.219.1709>.
- Reche Jr, A., Buffington, T.C., 1998. Increased tyrosine hydroxylase immunoreactivity in the locus coeruleus of cats with interstitial cystitis. *Am. Urol. Assoc.* 1045–1048.
- Sævik, B.K., Trangerud, C., Ottesen, N., Sørum, H., Eggertsdóttir, A.V., 2011. Causes of lower urinary tract disease in Norwegian cats. *J. Feline Med. Surg.* 13, 410–417. <https://doi.org/10.1016/j.jfms.2010.12.012>.
- Sparkes, A., 2018. Understanding feline idiopathic cystitis, 486 *Vet. Rec.* 182, 486. <https://doi.org/10.1136/vr.k1848>.
- Westropp, J.L., Delgado, M., Buffington, T.C., 2019. Chronic lower urinary tract signs in cats. *Vet. Clin. North Am. Small Anim. Pract.* 49, 187–209. <https://doi.org/10.1016/j.cvsm.2018.11.001>.
- Westropp, J.L., Welk, K.A., Buffington, T.C., 2003. Small adrenal glands in cats with feline interstitial cystitis. *J. Urol.* 170, 2494–2497. <https://doi.org/10.1097/01.ju.0000095566.63870.66>.